

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

Product identifier

Chemical Name	Mixture
Trade Names	Linchpin™
CAS No.	Mixture
EPA Identification Number	EPA Reg. No. 60063-xx

Relevant identified uses of the substance or mixture and uses advised against

Identified Use(s)	Herbicide / R&D Only
Uses Advised Against	It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

Details of the supplier of the safety data sheet

Company Identification	Sipcam Agro USA, Inc. 2525 Meridian Parkway Durham, NC 27713 United States of America
Telephone	(919) 226-1195

Emergency telephone number

Emergency Phone No.	CHEMTREC 24 hr. 1-800-424-9300 / 1 (703) 527-3887 (Collect calls accepted)
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SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

OSHA HCS (29 CFR 1910.1200)	Skin Sens. 1B; Repr. 1B; Eye Irrit. 2
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Label elements

Hazard Symbol(s)



Signal Word(s)

DANGER

Hazard Statement(s)

May cause an allergic skin reaction.
Suspected of damaging the unborn child.
Causes severe eye irritation.

Precautionary Statement(s)

Obtain special instructions before use.
Do not handle until all safety precautions have been read and understood.
Do not breathe dust.
Use only outdoors or in a well-ventilated area.
Wear protective gloves/protective clothing/eye protection/face protection.
Wash hands and exposed skin thoroughly after handling.
Contaminated work clothing should not be allowed out of the workplace.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists, get medical advice/attention.
IF ON SKIN: Take off immediately all contaminated clothing. Gently wash with

plenty of soap and water. If irritation (redness, rash, blistering) develops, get medical attention. Treat symptomatically. Contaminated clothing should be laundered before reuse.

IF exposed or concerned: Get medical advice/attention. Treat symptomatically.

Store in a well-ventilated place. Keep container tightly closed.

Store locked up.

Disposal should be in accordance with local, state or national legislation. Consult an accredited waste disposal contractor or the local authority for advice. Observe directions on label and instructions for use. See Also Section 15.

Other hazards

Additional Information

Very toxic to aquatic life. Very toxic to aquatic life with long lasting effects.

None

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Composition/information on ingredients	%W/W	CAS No.	Hazard Statement(s)
8-Quinolincarboxylic acid, 3,7-dichloro-	62.5	84087-01-4	Skin Sens. 1; H317
Mesotrione	8.7	104206-82-8	Aquatic Acute 1; H400 Aquatic Chronic 1; H410
Halosulfuron-methyl	4.1	100784-20-1	Repr. 1B; H360D Acute Tox 4; H302 Aquatic Acute 1; H400 Aquatic Chronic 1; H410
Sodium naphthalene-2-sulphonate	1.2	532-02-5	Skin Irrit. 2; H315 Eye Dam. 1; H318 Aquatic Acute 3; H402

Additional Information – None

SECTION 4: FIRST AID MEASURES



Description of first aid measures

Inhalation

Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. If breathing is laboured, administer oxygen. Treat symptomatically.

Skin Contact

Take off immediately all contaminated clothing. Gently wash with plenty of soap and water. If irritation (redness, rash, blistering) develops, get medical attention. Treat symptomatically. Contaminated clothing should be laundered before reuse.

Eye Contact

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists, get medical advice/attention.

Ingestion

Do not give anything by mouth to an unconscious person. Get medical advice/attention if you feel unwell. Rinse mouth.

Most important symptoms and effects, both acute and delayed

Fatal if inhaled. May cause an allergic skin reaction.

Indication of any immediate medical attention and special treatment needed

Call a POISON CENTER or doctor/physician if you feel unwell.

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

- Suitable Extinguishing Media
- Unsuitable Extinguishing Media

Extinguish with carbon dioxide, dry chemical, foam or water spray.
None known.

Special hazards arising from the substance or mixture

Combustion or thermal decomposition will evolve toxic and irritant vapours. Forms oxides of carbon and nitrogen, also chlorine and chlorinated compounds. Decontaminate equipment or materials used in pesticide fires.

Advice for fire-fighters

Fire fighters should wear complete protective clothing including self-contained breathing apparatus.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Wear protective gloves/protective clothing/eye protection/face protection. Wear suitable respiratory protective equipment if exposure to high levels of material are likely.

Environmental precautions

Do not allow to enter drains, sewers or watercourses.

Methods and material for containment and cleaning up

Sweep or shovel-up spillage and remove to a safe place. Transfer to a container for disposal or recovery.

Reference to other sections

None

Additional Information

None

SECTION 7: HANDLING AND STORAGE

Precautions for safe handling

Do not breathe dust. Use only outdoors or in a well-ventilated area. Wear protective gloves/protective clothing/eye protection/face protection. Wash hands and exposed skin thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace.

Conditions for safe storage, including any incompatibilities

- Storage temperature
- Incompatible materials

Store at room temperature. Do not allow material to freeze.
Strong oxidising agents. Substances that react with water.

Specific end use(s)

Herbicide

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Occupational Exposure Limits

SUBSTANCE.	CAS No.	(8hr TWA)		(STEL)		Note:
		PEL (OSHA)	TLV (ACGIH)	PEL (OSHA)	TLV (ACGIH)	
None	-----	-----	-----	-----	-----	-----

Exposure controls

Appropriate engineering controls

Avoid generation of dust.

Personal protection equipment

Eye/face protection



Wear protective eyewear (goggles, face shield, or safety glasses).

Skin protection (Hand protection/ Other)



Gloves.

Respiratory protection



Wear suitable respiratory protective equipment if exposure to high levels of material are likely.

Thermal hazards

Not normally required.

Environmental Exposure Controls

Do not allow to enter drains, sewers or watercourses.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance	Solid
Colour.	Not available.
Odour	Mild
Odour Threshold (ppm)	Not available.
pH (Value)	Not applicable.
Melting Point (°C) / Freezing Point (°C)	Not available.
Boiling point/boiling range (°C):	Not available.
Flash Point (°C)	Not applicable.
Evaporation rate	Not applicable.
Flammability (solid, gas)	Not available.
Explosive Limit Ranges	Not applicable.
Vapour pressure (Pascal)	Not available.
Vapour Density (Air=1)	Not available.
Density (g/ml)	Not available.
Solubility (Water)	Not available.
Solubility (Other)	Not available.
Partition Coefficient (n-Octanol/water)	Not available.
Auto Ignition Point (°C)	Not available.
Decomposition Temperature (°C)	Not available.
Kinematic Viscosity (cSt)	Not available.
Explosive properties	Not explosive.
Oxidising properties	Not oxidising.
Other information	Not available.

SECTION 10: STABILITY AND REACTIVITY

Reactivity	No hazardous reactions known if used for its intended purpose.
Chemical stability	Stable under normal conditions.
Possibility of hazardous reactions	None anticipated.
Conditions to avoid	Incompatible materials
Incompatible materials	Strong oxidising agents. Alkalis. Acids.

Hazardous decomposition product(s)

Combustion or thermal decomposition will evolve toxic and irritant vapours. Forms oxides of carbon and nitrogen, sulfur, also chlorine and chlorinated compounds. Decontaminate equipment or materials used in pesticide fires.

SECTION 11: TOXICOLOGICAL INFORMATION

Exposure routes: Inhalation, Skin Contact, Eye Contact

Information on toxicological effects

Substances

Not applicable

Mixtures

Acute toxicity	Not available.
Skin corrosion/irritation	Not available.
Serious eye damage/irritation	Not available.
Respiratory or skin sensitization	Not available.
Germ cell mutagenicity	Not available.
Carcinogenicity	Not available.
Reproductive toxicity	Not available.
STOT - single exposure	Not available.
STOT - repeated exposure	Not available.
Aspiration hazard	Not available.

Substances in preparations / mixtures

8-Quinolinecarboxylic acid, 3,7-dichloro- (CAS# 84087-01-4):

Acute toxicity	Oral: LD50 = 2190 mg/kg-bw (rat) Dermal: LD50 >2000 mg/kg-bw (rat)
Irritation/Corrosivity	Unlikely to cause eye irritation. Unlikely to cause skin irritation.
Sensitization	Not to be expected.
Repeated dose toxicity	Not to be expected.
Carcinogenicity	Not to be expected.

NTP	IARC	ACGIH	OSHA
No.	No.	No.	No.

Mutagenicity	Not to be expected.
Reproductive toxicity	Not to be expected.

Halosulfuron-methyl (CAS# 100784-20-1):

Acute toxicity (60% active)	Oral: LD50 = 1287 mg/kg-bw (rat) Dermal: LD50 >5000 mg/kg-bw (rat) Inhalation: LC50 >5.7 mg/L (4 hr) (rat) (Inhalable Dust)
Irritation/Corrosivity	Unlikely to cause eye irritation. Unlikely to cause skin irritation.
Sensitization	Not to be expected.
Repeated dose toxicity	Not to be expected
Carcinogenicity	Not to be expected.

NTP	IARC	ACGIH	OSHA
No.	No.	No.	No.

Mutagenicity	Not to be expected.
Reproductive toxicity	Suspected of damaging the unborn child.

Mesotrione (CAS# 104206-82-8):

Acute toxicity

Oral: LD50 >2000 mg/kg -bw
Dermal: LD50 >5000 mg/kg -bw
Inhalation: LC50 >2.08 mg/l (Aerosol), 4-hr. rat

Skin corrosion/irritation
Serious eye damage/irritation
Respiratory or skin sensitization
Germ cell mutagenicity
Carcinogenicity

Unlikely to cause skin irritation.
Unlikely to cause eye irritation.
It is not a skin sensitizer in animal tests.
Not to be expected.
Not to be expected.

NTP	IARC	ACGIH	OSHA
No.	No.	No.	No.

Reproductive toxicity
STOT - single exposure
STOT - repeated exposure
Aspiration hazard

Not to be expected. NOAEL = 71 mg/kg/day (mice)
Not to be expected.
Not to be expected. NOAEL = 71 mg/kg/day (mice)
Not available.

Additional Information

Not available.

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

Mesotrione (CAS# 104206-82-8):

Short term

LC50 (96 hour): >120 mg/L (fish)
EC50 (48 hour): 840 mg/L (*Daphnia magna*)
EC50 (7 days): 0.02 mg/L (Duckweed)

Long Term

EC10 (7 days): 0.0068 mg/L (Duckweed)

Halosulfuron-methyl (CAS# 100784-20-1):

Short term (60% active)

LC50 (96 hour): >100 mg/L(fish)
EC50 (48 hour): >100 mg/L (*Daphnia magna*)
EC50 (72 hour): 0.004 mg/l (*Selenastrum capricornutum*)

Long Term

Very toxic to aquatic life with long lasting effects.

Persistence and degradability

Part of the components are poorly biodegradable.

Bioaccumulative potential

Not determined.

Mobility in soil

Not determined.

Results of PBT and vPvB assessment

Not classified as PBT.

Other adverse effects

None known.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Disposal should be in accordance with local, state or national legislation. Consult an accredited waste disposal contractor or the local authority for advice. Observe directions on label and instructions for use. See Also Section 15.

Additional Information

None known.

SECTION 14: TRANSPORT INFORMATION

	Land transport (U.S. DOT) *	Sea transport (IMDG)	Air transport (ICAO/IATA)
UN number	Not classified as dangerous for transport. *	UN 3077	UN 3077
Proper Shipping Name		Environmentally hazardous substance, solid, n.o.s. (Mesotrione & Halosulfuron-methyl)	

Transport hazard class(es)	9	9
Packing group	III	III
Hazard label(s)	Marine Pollutant	Marine Pollutant
Environmental hazards	Yes	Yes
Special precautions for user	None Assigned	None Assigned

Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code: Not applicable

* For bulk shipments in U.S., can be shipped as UN3077, Environmentally hazardous substance, liquid, n.o.s. (Mesotrione & Halosulfuron-methyl), 9, PG III

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture:

TSCA (Toxic Substance Control Act) - Inventory Status: All components are listed and active, or exempt.

Designated Hazardous Substances and Reportable Quantities (40 CFR 302.4):

Chemical Name	CAS No.	Typical %wt.	RQ (Pounds)
None	----	-----	-----

SARA 311/312 - Hazard Categories: See SECTION 2 - HAZARDS IDENTIFICATION

SARA 313 - Toxic Chemicals (40 CFR 372):

Chemical Name	CAS No.	Typical %wt.
Chlorothalonil	1897-45-6	~ 38

SARA 302 - Extremely Hazardous Substances(40 CFR 355):

Chemical Name	CAS No.	Typical %wt.	RQ (Pounds)	TPQ (Pounds)
None	----	-----	-----	

Proposition 65 (California):

Chemical Name	CAS No.	Typical %wt.	Hazards
Chlorothalonil	1897-45-6	~ 38	Cancer

SECTION 16: OTHER INFORMATION

The following sections contain revisions or new statements: 1-16

Date of preparation: May 12, 2023

Hazard Statement(s) and Risk Phrases Listed in: SECTION 2:/ SECTION 3:

Hazard Statement(s)

- H302: Harmful if swallowed.
- H315: Causes skin irritation.
- H317: May cause an allergic skin reaction.
- H318: Causes serious eye damage.
- H360D: May damage the unborn child.
- H400: Very toxic to aquatic life.
- H401: Toxic to aquatic life.
- H402: Harmful to aquatic life.
- H410: Very toxic to aquatic life with long lasting effects.

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